GURDHARAM JEET SINGH

Full-Stack Software Engineer • Web Systems & Cloud Automation Architect

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer and Cloud Automation Architect with proven production experience building high-performance React, Next.js, and TypeScript web applications and scalable distributed systems. Shipped commercial platforms including SOLUM Minerals, Branders, and Dream Heights with 99+ Lighthouse scores and fluid micro-interactions. Architect of 24/7 cloud automation fleets on GitHub Actions, low-latency telephony event systems (<700ms turn latency), and Redis/BullMQ distributed message pipelines processing 25,000+ daily events. Available immediately for remote software engineering internships, co-ops, and full-time roles worldwide.

TECHNICAL SKILLS

Frontend & Web Systems: React, Next.js (App Router, SSR, RSC), TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, WebSockets, Three.js, WebGL2 (Camera Kit), GLSL Shaders, Lenis Smooth Scroll, Technical SEO, GEO, JSON-LD Schema.

Backend & Distributed Architecture: Node.js, Express, Python (FastAPI, Asyncio), Go, PostgreSQL, Redis & BullMQ Queues, RESTful APIs, Webhook Orchestration, Docker, Linux CLI / Bash, systemd, Tailscale, Headless Playwright / Puppeteer.

Cloud, DevOps & AI Workflows: GitHub Actions CI/CD, AWS (S3, Lambda), GCP, Railway, Sarvam AI Indic LLMs, Groq LPU, Voice AI, SIP, VoIP, Silero VAD, RAG, Model Context Protocol (MCP), FFmpeg Audio/Video Automation, TensorFlow Lite INT8, Flutter Isolates.

Testing, Quality & Web Performance: Playwright & Puppeteer E2E Testing, Jest, React Testing Library, Core Web Vitals Optimization (LCP, CLS <0.02), Lighthouse 99+ Audits, Accessibility (WCAG 2.1, WAI-ARIA), Git & GitHub Actions CI/CD, Vite, Postman, RESTful API Design.

KEY ENGINEERING PROJECTS

Autonomous 5-Channel YouTube Media Fleet (yt-auto) — Cloud / GitHub Actions | 2025 – Present

Lead Automation Architect • Open Source (github.com/gurination1/yt-auto)

- **100% Headless Cloud Fleet Execution:** Engineered resilient 5-channel media generation fleet running autonomously on GitHub Actions across 15 daily slots with zero local machine dependencies.
- **Dual-Gate Multimodal Quality Control:** Integrated FFmpeg black-frame detector (`blackdetect=d=0.8`) and Gemini Multimodal Judge AI (evaluating narrative flow, pacing, and subtitle sync) enforcing a strict ≥85/100 publication threshold.
- **Algorithmic Audio & Video Rendering:** Built headless audio ducking and dynamic background scoring pipeline in Python/FFmpeg with millisecond word-level subtitle alignment (`.ass` dynamic styling).

SOLUM Minerals, Branders & Dream Heights Web Suite — Remote / Production | 2025 – Present

Full-Stack Web Architect & Systems Developer

- **SOLUM Minerals Platform:** Architected low-carbon mineral geology web application using Next.js 15 (App Router, Server Components), TypeScript, and Tailwind CSS. Built custom responsive components, dynamic product showcases, and interactive data visualizers with 99+ Lighthouse performance rating.
- **Branders Digital Experience Platform:** Engineered high-performance brand agency web platform with modern interactive components, fluid micro-interactions, zero layout shifts (CLS < 0.02), and physics-hardened mobile navigation.
- **Dream Heights Real Estate Platform:** Architected real estate portal with spatial property filtering, virtual floor plan previews, interactive image carousels, and automated client inquiry pipelines.
- **Mobile & Edge Systems:** Built Fasal Doctor (plant pathology Flutter / TensorFlow Lite INT8) and DoodHisaab (dairy ledger system with Flutter Isolates).

Genesis Coding Agent & AST Self-Healing Engine — Autonomous Systems / Python | 2025 – Present

Lead Architect • AST Code Synthesis & Cloud Workflows

- **Autonomous AST Mutation Loops:** Engineered self-healing agent engine that parses Python Abstract Syntax Trees (AST), generates surgical code diffs, and validates test suites with zero manual intervention.
- **Playwright & Headless Cloud Integration:** Implemented end-to-end browser verification harnesses, capturing multi-surface DOM state and generating automated visual validation logs.

WORK EXPERIENCE

Gurdharam (Independent Software Consultancy) — Remote / Punjab, India | 2025 – Present

Full-Stack Web & Autonomous Systems Engineer

- **Telephonic Voice AI & Speech Agent Architecture:** Architected real-time telephony voice agent over Exotel WebSockets, SIP, and VoIP, achieving <700ms turn latency across Sarvam AI Indic models and Groq LPUs with Silero VAD barge-in for Neo and Priya conversational personae.
- **High-Throughput Webhook & Messaging Gateway:** Engineered automated outreach and webhook triage gateway using Official WhatsApp APIs, Redis, and BullMQ, reliably processing 25,000+ incoming webhooks with sub-5-minute SLA.
- **Web AR Lens Pipeline (SnapAR Studio):** Developed WebGL2 Camera Kit testing portal integrating custom GLSL shaders, 3D asset conversion pipelines, and automated package verification.

EDUCATION

Baba Farid Group of Institutions (BFGI) — Punjab, India | Expected Graduation: May 2027

Bachelor of Science in Computer Science & Software Systems

- **Core Curriculum:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering.

CERTIFICATIONS & REGISTRATIONS

- **Govt. MSME Registered Enterprise:** Ministry of Micro, Small and Medium Enterprises (UDYAM-PB-03-0079744).
- **Developer Partnerships & Verification:** Sarvam AI Startup Developer Member (Priority Indic LLM rates) • Meta Verified business entity.